Flagness and Chain-Count Invariants of Combinatorial Barycentric Subdivision

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Abstract

For every partially ordered set, its nonempty finite chains form an actual abstract simplicial complex. We prove this complex is flag, its d-dimensional face count is the number of chains with $d+1$ elements, and these counts are invariant under order isomorphisms. Applying this construction to a polytope's nonempty face poset is the standard *combinatorial* barycentric subdivision. We explicitly distinguish the whole polytope from its boundary and use actual face subsets, without a simplicial-input restriction. The source's unspecified invariant is interpreted as the enumerative f-vector, not a complete classifier or a quantity preserved by performing subdivision. No formal geometric triangulation theorem is claimed.

1 Source Scope and Conventions

The exact English source asserts the flag property of barycentric subdivision and describes its invariant as chain counting of the original faces. The Chinese source similarly refers to its combinatorial type, flagness and face-chain counts. Neither defines a particular invariant or says that numeric counts alone classify the subdivision. We prove the explicitly declared standard combinatorial interpretation: the entire chain-incidence family describes the abstract subdivision, and its f-vector is obtained by counting chains by their number of elements. This vector is invariant under combinatorial isomorphism; it is not asserted unchanged by subdivision.

The domain is nonempty bounded convex polytopes. For a polytope Q, let $\mathcal{F}(Q)$ be its actual finite face family, with faces regarded as *subsets* of the ambient affine space. Define

$$P_{\text{full}}(Q) = \{F \in \mathcal{F}(Q) : F \neq \varnothing\}, \qquad P_{\partial}(Q) = \{F \in \mathcal{F}(Q) : F \neq \varnothing,\ F \neq Q\},$$

both ordered by inclusion. The full convention includes Q; the boundary convention does not. Empty faces are not subdivision vertices. In particular, the boundary of a point has no nonempty faces. The combinatorial subdivision is *defined* to be the order complex of the corresponding face poset. This standard description also applies to nonsimplicial polytopes [1, p. 11]; more generally, the face-poset order complex of a polyhedral complex gives its combinatorial barycentric subdivision [2, Section 2].

Frozen source provenance:

```
Commit:  b355e42906e6eaf52bc7e9d005e0ef02d8aa7400
Blob:  6c54d99d7caf701ff7e55c4530c2eb441a049aec.
```

Both complete source languages are retained in the package's source document.

2 The Actual Order Complex is Flag

Let P be any partially ordered set, not necessarily finite. A strict finite chain means a finite subset totally ordered by $\leq$. Repetitions are absent because it is a set. Chains need not be saturated and may skip ranks. The length of a chain means its number of elements, not its number of inequalities. Define the abstract complex

$$\Delta(P) = \{s \subseteq P : s \text{ finite, nonempty, and a chain}\}.$$

Every nonempty subset of a chain is a chain, and every singleton is a chain. These facts prove the actual Mathlib abstract-complex structure fields. The empty set is not a stored face under Mathlib's convention.

The comparability graph has vertex set P and adjacency

$$x \sim y \quad \Longleftrightarrow \quad x \neq y \ \text{ and } \ (x \leq y \text{ or } y \leq x).$$

It is an actual simple graph: adjacency is symmetric and irreflexive. Its edges are exactly the two-element faces of $\Delta(P)$. A finite set is a clique if and only if it is a chain, and therefore

$$s \in \Delta(P) \quad \Longleftrightarrow \quad s \neq \varnothing \ \text{ and } \ s \text{ is a clique in the comparability graph.}$$

Theorem 1. *For every poset P, the abstract order complex $\Delta(P)$ is flag.*

Proof. Let s be a nonempty finite vertex set such that every distinct pair in s is an edge. Such pairs are comparable. Equal pairs are comparable by reflexivity. Thus s is a chain and is a face of $\Delta(P)$. This proves the clique-to-face condition, with no assumed flagness or closure predicate. □

3 Chain Counts and the Correct Invariant

Assume P is finite. Put

$$\mathcal{S}_k(P) = \{s \in \Delta(P) : |s| = k\}, \qquad \mathcal{C}_k(P) = \{s \subseteq P : s \neq \varnothing,\ s \text{ a chain},\ |s| = k\}.$$

Membership in the actual complex gives an actual bijection $\mathcal{S}_k(P) \simeq \mathcal{C}_k(P)$, by sending the finite vertex set to the same underlying finite chain, with inverse inclusion as a face. Both directions are proved; no counting identity is an external hypothesis. By definition a d-dimensional abstract simplex has $d+1$ vertices, so

$$f_d(\Delta(P)) = |\mathcal{S}_{d+1}(P)| = |\mathcal{C}_{d+1}(P)|.$$

For empty P, all these nonempty-face counts are zero.

Theorem 2. *If $e : P \simeq Q$ is an order isomorphism of finite posets, then $f_d(\Delta(P)) = f_d(\Delta(Q))$ for every $d \geq 0$.*

Proof. An order isomorphism preserves and reflects comparability, so the finite image $e(s)$ is a chain if and only if s is. Injectivity preserves its cardinality, and nonemptiness is preserved. Taking finite images under e and e^{-1} gives mutually inverse maps of the actual face sets with each cardinality. The resulting bijection preserves every f_d. □

The whole chain-incidence complex contains more data than the numeric count vector. No theorem here states that the latter determines its isomorphism class, reconstructs the original polytope, or is equal to its pre-subdivision f-vector. For example a square boundary subdivides into eight vertices and eight edges, while the full square subdivides into face counts $(9, 16, 8)$. These examples clarify the two conventions; they are not the universal proof.

4 Actual Face Families and Formal Scope

Lean proves the above results for every poset and every finite family $\mathcal{A}$ of *actual subsets* of any ambient type E. It filters out empty sets, uses actual subset inclusion, and constructs the order complex. The boundary constructor additionally removes the specified whole face. The flagness and f-vector chain formulas follow for both conventions. An arbitrary input family need not itself be polytopal: the universal theorem is stronger, and specializes to the actual $\mathcal{F}(Q)$ of a polytope. No downward closure under arbitrary original *vertex* subsets is required. Thus a quadrilateral original face is one subdivision vertex, not a tetrahedron.

The standard combinatorial face-poset definition and the finiteness of the actual face family of a bounded polytope identify the specialization in ordinary mathematics. The Lean statements are explicitly universal finite-family statements, not theorems about a separate geometric Polytope type. No barycenter coordinates, affine independence, geometric gluing or coverage are advertised as formally constructed. Such stronger geometric claims would require further bridges and are not obtained merely by giving an arbitrary family a geometric name.

The core consists of 21 public theorems, plus the proved face and transport equivalence definitions. Actual source replay and every public theorem's compiled type and axiom footprint pass on Lean 4.33.1 with Mathlib commit:

```
0df444a360eaa60ab8c11dca51a86af692955474.
```

Only `propext`, `Classical.choice` and `Quot.sound` occur. The portable package reproducer checks all nine physical Git dependencies, their pinned origins/HEADs, tracked raw bytes and index/tree agreement before and after execution. It cleans only this project's build outputs, actually builds Main, replays the mathematical source and Main with warnings as errors, and checks all 21 compiled theorem types and axiom footprints. Its finite face incidence examples are regression checks, not the general proof. Installed Lean and dependency build caches remain trust inputs, not an empty-machine bootstrap, clean build of all Mathlib dependencies or formal geometric triangulation theorem. The separate AI-agent review record does not imply human expertise, organizer acceptance, first-solver priority or an award.

References

[1] F. Brenti and V. Welker, *f-vectors of barycentric subdivisions*. Author-hosted manuscript, September 2007, especially p. 11.

[2] J. A. Olarte and F. Santos, *Hypersimplicial subdivisions*. Selecta Mathematica **28** (2022), article 4 (online 2021). DOI: `10.1007/s00029-021-00722-6`, Section 2.